

Definitions – and their misuse

OBJECTIVE

Certain terms have a specific meaning when used within the context of the Wiring Regulations and in some circumstances, because their exact meaning is not fully understood, the requirements of BS 7671 are not fully satisfied. The purpose of this article is to remind contractors not to overlook the Definitions provided in Part 2 of BS 7671.

Introduction

In some circumstances, ambiguity regarding the meaning of a particular term when used in the context of BS 7671 can result in a requirement not being fully satisfied, or, conversely, it can lead to work such as protective bonding being carried out unnecessarily. Therefore, to support the user, the specific meaning of a range of terms used within BS 7671 is provided in Chapter 2 of BS 7671, 'Definitions'; that is, when a definition is given in BS 7671 it has that meaning and no other. This article discusses some of the terms that consistently occur in enquiries to NICEIC and ELECSA.

Departure

A definition for 'departure' was not included previously in BS 7671, but is included in Part 2 of the 18th edition, and those responsible for the design, installation and certification of electrical installations should ensure that where a departure is used, it is appropriate.

The requirements contained in BS 7671 are not intended to restrict or obstruct the design of an electrical installation, and so, where required, it is acceptable for the designer, in agreement with other parties, to use a 'departure' (Regulation 120.3). The designer¹,

who is responsible for all aspects of the design, such as the selection of protective devices, cables and equipment may, for example, select technology or equipment that is not yet covered by a British or Harmonised Standard. However, in taking the deliberate decision not to comply fully with BS 7671, the designer is responsible for confirming that the departure:

- results in no lesser degree of safety than that obtained by full compliance with BS 7671, and
- is recorded on the appropriate electrical certificate (Regulations 120.3, 133.1.3 and 133.5 refer).

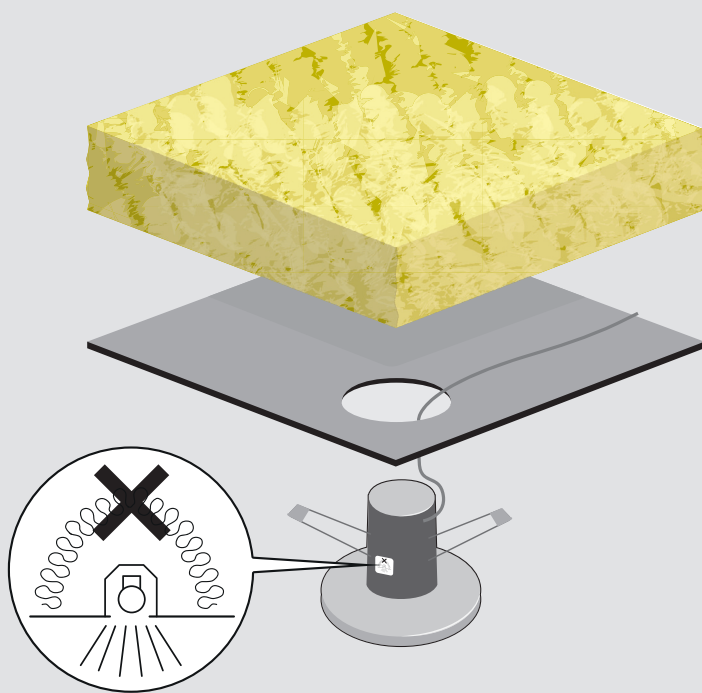
Departures and certification

For certification purposes it is important to distinguish between a departure and a 'non-compliance'. For example, where the designer specifies a cable that is not covered by a product standard listed in BS 7671, such as 'SY' cable, and the use of the cable provides the same level of safety as that required by BS 7671, then it is permitted to record the use of the cable as a departure on the electrical certificate. However, it would not be acceptable to do so if the installation of the 'SY' cable resulted in a reduced level of safety. This would be the case, for example, if the 'SY' cable were to be installed in the ground, without any additional mechanical protection being provided – particularly as the metallic braid of an 'SY' cable is not suitable for use as a protective conductor (Regulation 522.8.10 refers).

Sometimes, as a result of design changes made during installation work, a 'non-compliance' might be incorrectly recorded as a 'departure'. In Fig 1, for example, the original design may have specified a recessed downlighter that is not suitable to be covered by thermal insulation but, as a result of a change made during the installation, it may end up being installed in a position where it is covered by thermal insulation. However, this is a 'non-compliance', not a 'departure'. As required

¹ Where there is no designated 'designer' for the electrical installation, as is commonly the case for domestic rewires and similar works, it is the electrical contractor undertaking the installation work who will inevitably assume the role of the designer.

Fig 1 Luminaire not suitable for covering with thermally insulating material – Table 55.3 of BS 7671



by Regulation 644.1.1, this omission must be corrected before the electrical certificate is issued, such as by fitting a luminaire that is suitable or implementing additional measures to prevent the risk of fire (Regulation 421.1.2 refers).

So that departures are used appropriately, NICEIC and ELECSA certificates require that, in addition to recording the presence of departures, full details of each departure and the designer's intent is provided on an additional numbered page and appended to the certificate.

Skilled person (electrically)

As stated in the definition, Regulation 16 of the EWR 1989 requires persons to be competent to prevent danger or injury. For these reasons, the term 'skilled person' is used in BS 7671 to identify that an appropriate level of electrical

competence is required. Indeed, the term 'electrically' is implicit to this definition, and so is not included where the definition is used within BS 7671.

For example, BS 7671 requires that inspection and testing of electrical installations is undertaken by skilled persons competent in such work (Regulations 641.6 and 651.5 refer). NICEIC and ELECSA consider that, to undertake the inspection and testing of an electrical installation, persons should, as a minimum:

- have sufficient knowledge and experience of electrical installation matters to avoid danger to themselves and to others, and
- be familiar with, and understand, the requirements of BS 7671, including those relating to inspection, testing, certification and reporting, and
- have a sound knowledge of the particular type of installation to be inspected and tested, and
- have sufficient information about the function and construction of the installation to allow them to proceed in safety.

Consider as an example where Regulation 411.3.3 requires additional protection to be provided by an RCD with a rated residual operating current ($I_{\Delta n}$) not exceeding 30 mA, for socket-outlets rated at 32 A or less. Although an exception to this requirement is permitted for an installation other than a dwelling, it must be based on a documented risk assessment. The purpose of the assessment is to determine whether omitting an RCD will result in reducing the safety of the installation. For these reasons, BS 7671 requires an appropriately **skilled person** to be involved in making such a decision (Appendix 2, item 11, of BS 7671 refers). Where an exception to Regulation 411.3.3 is used, it should be identified on the certificate, as shown in Fig 2, and the risk assessment (or a copy of it) must be appended to the certificate.

Extraneous-conductive-part

An extraneous-conductive-part is a conductive part that does not form part of the electrical installation, but is liable to introduce a potential – generally Earth potential. Due to the risk

Fig 2 Luminaire not suitable for covering with thermally insulating material – Table 55.3 of BS 7671

PART 4 : DECLARATION FOR THE ELECTRICAL INSTALLATION WORK (this option may be used where the design, construction, inspection & testing have been the responsibility of one person)

DESIGN, CONSTRUCTION, INSPECTION & TESTING (The extent of liability of the signatories is limited to the work detailed in PART 2)

I, being the person responsible for the design, construction, inspection and testing of the electrical installation, particulars of which are described in PART 2, having exercised reasonable skill and care when carrying out the design and additionally where this certificate applies to an addition or alteration, having confirmed that the safety of the existing installation is not impaired, hereby CERTIFY that the design, construction, inspection and testing for which I have been responsible is to the best of my knowledge and belief in accordance with BS 7671: 2018, amended to (date) except for the departures, if any, detailed on attached page(s) (N/A.) (Regulations 120.3, 133.1.3 and 133.5).

• Permitted exception applied (411.3.3: Yes/NA

Risk assessment attached: (✓/.....)

Page No(s) (10/.....)

• Where selectivity is required, details of the verification appended (536.4): (N/A.) Page No(s) (N/A.)

of dangerous potentials existing within an installation, Regulation 411.3.1.2 requires that all extraneous-conductive-parts within each installation, such as structural steelwork and metallic services, are connected to the MET using protective bonding conductors complying with Chapter 54 of BS 7671. However, as it is now commonplace for installation services to be formed of non-conductive material, care needs to be exercised to avoid installing protective bonding conductors unnecessarily.

It should be noted that an item that is conductive may not necessarily introduce a potential. In order to present a risk of electric shock, the conductive part must be at Earth potential (that is, in contact with the general mass of the Earth) and, crucially, must also be accessible.

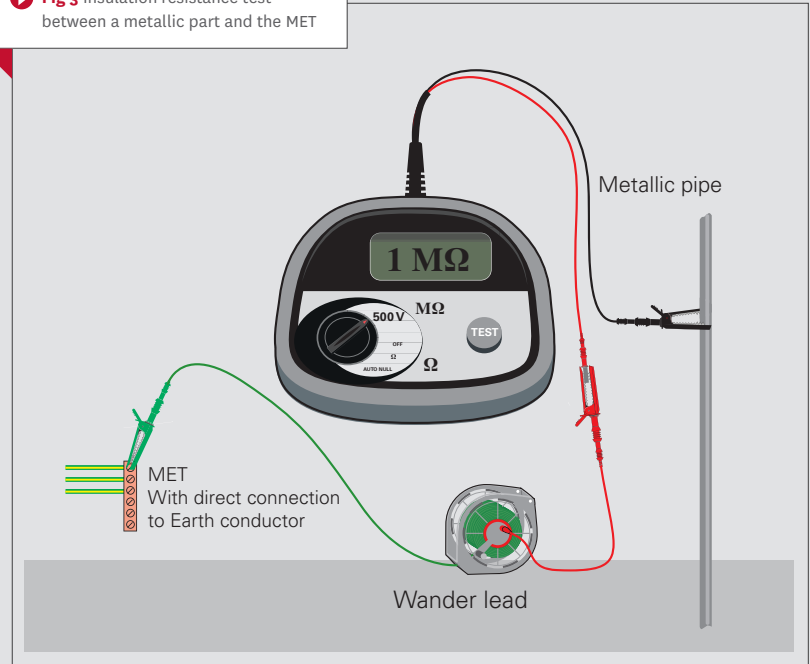
A conductive part, such as a metal duct or pipe, that is covered with, or enclosed by, insulating material, or an item that is out of reach and not in contact with any exposed-conductive-part or extraneous-conductive-part, is not liable to introduce a potential. Likewise, a metallic pipe having an insulating section at point of entry into a building is not deemed an extraneous-conductive-part (Regulation 411.3.1.2 refers).

For a new installation, any doubt regarding protective bonding should be referred to the installation designer for consideration and confirmation as to the requirement for main, and where applicable, supplementary bonding. In the case of periodic inspection, the inspector should make an assessment based on all the relevant circumstances. One method of verifying whether a conductive part requires bonding is to carry out insulation resistance testing at 500 V DC between the MET and the metallic part, or pipework, as shown in Fig 3.

Where the measured value of insulation resistance between the item and the MET is such that it would prevent any dangerous level of current flow between the part and Earth, the item may be considered to be an isolated metallic part, rather than an extraneous-conductive-part. Based on the 30 mA (RCD) current value (the level of current that BS 7671 deems a shock risk), and a body resistance of 1 000 Ω for a 230 V (50 Hz) supply, the metallic part would not require connecting to the MET where the measured value of insulation resistance 'IR' is greater than:

$$\text{Minimum } IR = \frac{U_o}{I_{\Delta n}} - R_{body} = \frac{230}{0.03} - 1\,000 = 6.67 \text{ k}\Omega$$

Fig 3 Insulation resistance test between a metallic part and the MET



The value of 1 000 Ω for the body resistance (R_{body}) is chosen from PD IEC/TS 60479-1 (based on hand-to-hand contact in dry conditions).

Note: In some circumstances it may be appropriate to use a lower current value than 30 mA, typically 10 mA, for the calculation. This may be appropriate where, for example, circuits are not protected by 30 mA RCDs. Where based on the assumed risk assessment, for example, the inspector chooses 10 mA as an assumed value, the minimum IR would be 22 kΩ (below).

$$\text{Minimum } IR = \frac{U_o}{I_{\Delta n}} - R_{body} = \frac{230}{0.01} - 1\,000 = 22 \text{ k}\Omega$$

Conclusion

The purpose of the definitions provided in Part 2 of BS 7671 is to support those referencing BS 7671 by providing a clarification of the specific meaning of certain terms when used within the context of the Wiring Regulations. For these reasons, it is important that the definitions provided are not overlooked when doubt exists to the technical intent of a particular regulation. Ⓢ